

VIVEKANAND EDUCATION SOCIETY'S INSTITUTE OF TECHNOLOGY

(An Autonomous Institute Affiliated to University of Mumbai
Department of Computer Engineering)

Department of Computer Engineering



Project Report
on

AutoGenie: Agentic AI Agent for Video Pre-Production Planning & Asset Sourcing

In collaboration with



Helmer: AI-native platform supercharging video
creation for brands and creators.

In partial fulfillment of the Fourth Year (Semester–VII), Bachelor of
Engineering

(B.E.) Degree in Computer Engineering at the University of Mumbai
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(2025-26)

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CERTIFICATE of Approval

This is to certify that _____ of Fourth Year Computer Engineering studying under the University of Mumbai has satisfactorily presented the project on “*AutoGenie: Agentic AI Agent for Video Pre-Production Planning & Asset Sourcing*” as a part of the coursework of PROJECT-I for Semester-VII under the guidance of *Dr. Sharmila Sengupta* in the year 2025-2026.

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Computer Engineering Department

COURSE OUTCOMES FOR B.E PROJECT

Learners will be to:-

Course Outcome	Description of the Course Outcome
CO 1	Do literature survey/industrial visit and identify the problem of the selected project topic.
CO2	Apply basic engineering fundamental in the domain of practical applications FOR problem identification, formulation and solution
CO 3	Attempt & Design a problem solution in a right approach to complex problems
CO 4	Cultivate the habit of working in a team
CO 5	Correlate the theoretical and experimental/simulations results and draw the proper inferences
CO 6	Demonstrate the knowledge, skills and attitudes of a professional engineer & Prepare report as per the standard guidelines.

ABSTRACT

This project introduces an AI Agent for Video Pre-Production Planning and Asset Sourcing, designed to streamline the creative workflow for content teams. Traditional pre-production processes are time-intensive, involving manual transformation of briefs into structured plans, asset gathering, and ensuring license compliance, often leading to inefficiencies and creative bottlenecks. The proposed agentic AI system addresses these challenges by leveraging autonomous, collaborative agents that unify prompt interpretation, video classification, planning, and asset management into a cohesive, iterative platform.

At its core, the system parses user briefs to classify video types, generate shot/asset plans with timing constraints, and decide per-asset whether to source from the web (via search, crawling, filtering, and licensing checks) or generate anew (using model APIs for images, videos, and audio). A conversational chat interface enables rapid refinements with versioned diffs, while modular tools ensure coherence, precision, and legal safety. Built on a React frontend, LLM-orchestrated backend, and data layers like Postgres and vector stores, the platform outputs production kits including storyboards, asset inventories, and exports.

By automating and democratizing pre-production, this agentic solution enhances efficiency, creativity, and accessibility for creators, reducing turnaround times and fostering scalable video content development without requiring deep technical expertise.

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Chapter 1 : Introduction

1.1 Introduction:

Artificial Intelligence (AI) has become a transformative force across industries, enabling innovative applications in education, healthcare, finance, entertainment, and more. However, developing AI-driven systems remains a complex, resource-intensive process, requiring expertise in model integration, data handling, multimodal processing, and deployment. These challenges create significant barriers for individuals, startups, and organizations seeking to leverage AI for creative and practical solutions.

To address this gap, we propose an Agentic AI Platform tailored for video pre-production planning and asset sourcing, designed to streamline content creation workflows. The platform empowers users to transform creative briefs into structured, license-safe video plans with minimal technical expertise. By leveraging autonomous, collaborative AI agents, it unifies prompt interpretation, video type classification, shot/asset planning, and automated sourcing or generation of assets (images, videos, audio) via web crawling, license filtering, or model APIs. A conversational interface facilitates iterative refinements with versioned diffs, ensuring transparency and efficiency in plan evolution. The system outputs comprehensive production kits, including storyboards, asset inventories, and exportable formats like PDF and JSON.

Built on a modular architecture with a React frontend, LLM-orchestrated backend, and robust data layers (Postgres, vector stores), the platform ensures scalability, consistency, and legal compliance. By abstracting complexities into intuitive, reusable components, this solution democratizes AI-driven video pre-production, enabling creators to produce high-quality content efficiently and fostering innovation across diverse use cases.

1.2 Motivation

The rapid advancement of Artificial Intelligence (AI) has revolutionized numerous industries by automating complex processes and enhancing creative workflows. In the field of digital content creation, particularly video production, there remains a significant challenge in efficiently managing the pre-production phase, which often involves extensive planning, asset sourcing, and coordination among multiple stakeholders. Current tools are either too technical or fragmented, making it difficult for creators, especially those with limited technical expertise, to seamlessly translate creative ideas into structured production plans. This creates a pressing need for an intelligent system that can autonomously assist users in organizing video projects, sourcing assets safely within license constraints, and facilitating iterative collaboration. The motivation behind this project stems from the desire to simplify and accelerate video pre-production workflows through an agentic AI platform, empowering creators to focus more on creativity and less on logistical complexities. This solution aims to bridge the gap between conceptualization and execution by leveraging AI's capabilities to

automate planning, asset gathering, and production kit generation, making video production more accessible, efficient, and legally compliant.

1.3 Problem Definition:

Content creation teams face significant challenges in the pre-production phase of video development, where transforming a creative brief into a structured, license-compliant plan demands excessive time and resources. The process involves multiple complex tasks: classifying the video format, defining shot sequences and asset requirements with precise timing, deciding whether to source assets from external repositories or generate them anew, and ensuring all assets align with brand identity and legal licensing requirements. These tasks are often manual, fragmented, and prone to inefficiencies, leading to delays and increased costs. Additionally, iterating on plans requires cumbersome communication, and sourcing or generating assets lacks streamlined automation, further complicating the workflow.

To address these challenges, there is a need for an AI-driven agent that unifies and automates the pre-production process. This agent should interpret user prompts to classify video types and enforce format constraints, generate detailed shot and asset plans, make intelligent source-versus-generate decisions for each asset, and provide automated tools for web-based asset sourcing (with search, crawling, deduplication, and licensing filters) and safe asset generation (via model APIs for images, videos, and audio). A conversational interface is essential for iterative plan refinement with clear versioned diffs, enabling rapid convergence on a final plan. The solution must deliver a comprehensive production kit, including a storyboard PDF, CSV/JSON asset lists, prompts, and links, while ensuring quality through coverage, coherence, retrieval precision, legal compliance, and user satisfaction. By streamlining these processes, the agent aims to reduce pre-production time, enhance creative efficiency, and make video content creation accessible to users with minimal technical expertise.

1.4 Relevance of the Project:

The proposed project, *AutoGenie: Agentic AI Agent for Video Pre-Production Planning & Asset Sourcing*, holds significant relevance in today's rapidly evolving digital media landscape, where video content has become a dominant mode of communication, marketing, and storytelling. With the increasing demand for high-quality and visually compelling videos across industries—ranging from entertainment and education to advertising and social media—creative teams often face challenges in efficiently managing the pre-production phase. This stage involves multiple complex, time-consuming tasks such as interpreting creative briefs, planning shot sequences, sourcing or generating assets, and ensuring copyright compliance. These tasks require extensive coordination, domain expertise, and manual effort, often resulting in bottlenecks, inconsistent quality, and higher production costs.

AutoGenie addresses these inefficiencies by introducing an agentic AI system capable of autonomously understanding creative briefs, generating structured video plans, and sourcing

or generating media assets with built-in legal and licensing compliance. The relevance of the project lies in its ability to streamline and democratize the pre-production process through AI-driven automation and collaboration. By leveraging technologies such as large language models (LLMs), autonomous multi-agent frameworks, and intelligent retrieval systems, the platform empowers creators, marketing professionals, and production teams to rapidly convert ideas into production-ready plans—without requiring deep technical knowledge or specialized tools.

Furthermore, in the context of the global creative industry’s shift toward AI-assisted workflows, *AutoGenie* exemplifies how human-AI collaboration can enhance both productivity and creativity. Its conversational interface fosters iterative refinement and transparency, making the planning process interactive, intuitive, and accessible. The system’s modular architecture also ensures adaptability to diverse use cases, including corporate marketing, educational content creation, and film pre-visualization. Thus, this project is not only relevant as a technological innovation but also as a practical enabler of efficiency, scalability, and inclusivity in the creative ecosystem—paving the way for the next generation of intelligent, collaborative content production tools.

1.5 Methodology used:

In this system, an **LLM-based Agent Orchestrator** serves as the central controller, managing planning, tool execution, and iterative improvements across multiple interconnected agents. The orchestrator coordinates with a **Planner Agent**, which defines the asset list and determines the workflow required for generating or sourcing digital assets. Decision-making is handled by the **Source/Generate Decider Agent**, which selects whether an asset should be generated using AI models or retrieved from existing sources based on parameters such as license validity, relevance, and quality. Two key agents execute these decisions — the **Generation Agent**, which creates assets through API calls to AI vendors like OpenAI or Runway using predefined prompt templates, and the **Searching Agent**, which gathers data through web searches and APIs, extracting metadata such as titles, tags, durations, and license details. The searching agent further applies license filtering and semantic scoring to ensure compliance and contextual accuracy.

After asset creation or retrieval, the data undergoes **embedding generation and deduplication**, where near-duplicates and low-quality items are filtered out to maintain dataset precision. Embeddings and semantic scores are stored in a **Vector Database**, while finalized media outputs such as thumbnails, previews, and videos are stored in an **Object Store**. The structured metadata from both generation and search pipelines is stored in a **PostgreSQL database**, ensuring centralized data management. Finally, the **Production Kit Generator** compiles this structured data into organized formats such as storyboards, PDFs, or JSON files for production use. This architecture demonstrates a well-orchestrated, intelligent pipeline that automates content planning, generation, validation, and production streamlining the end-to-end workflow for scalable and efficient media generation.

Chapter 2: Literature Survey

2.1 Research Papers referred:

Year	Objective	Methodology	Advantages	Limitations
2025 [1]	Develop an automatic film generation system combining cinematic principles with generative AI to enhance storytelling and content creation.	Uses prompt-based generative models (e.g., diffusion and transformer-based architectures) trained on cinematic datasets to simulate lighting, framing, and narrative coherence. Incorporates film grammar principles (shot composition, camera movement) into AI outputs.	Provides a fusion of creativity and automation , enabling faster pre-production workflows and lower production costs. Enhances visual realism and narrative cohesion in AI-generated films.	Limited contextual control over story arcs and emotional tone. Struggles with consistent character continuity across scenes and scalability for feature-length productions.
2025 [2]	Create a multi-agent framework for collaborative film generation integrating director, editor, and cinematographer agents powered by diffusion models .	Utilizes a multi-agent reinforcement learning setup where each agent specializes in creative domains (storyboarding, cinematography, editing). Agents communicate through a coordination module for decision-making.	Promotes collaborative creativity and diversity of output. Reduces creative bias by integrating multiple AI “roles.” Can simulate professional film crew dynamics for faster concept generation.	Lacks detailed implementation for real-time coordination and synchronization between agents. Computationally expensive due to multiple model interactions.
2025 [3]	Conduct a literature review on recent advancements in generative AI for film creation	Systematic literature review of research papers, patents, and case studies from 2018–2025, covering diffusion models, GANs, and multimodal LLMs in video	Offers a comprehensive overview of trends and technologies, helping researchers identify research gaps in narrative-based AI generation.	Many papers reviewed are theoretical, lacking practical validation . Limited coverage of ethical and copyright implications in automated content creation.

		synthesis.		
2025 [4]	Provide a comparative analysis of current AI-driven cinematic tools for pre-production and post-production workflows	Mixed-method review combining qualitative content analysis and quantitative performance benchmarking of AI tools (e.g., Runway, Pika Labs, Sora).	Enables practitioners to select suitable tools for specific film processes. Presents quantitative performance data on model output quality and time efficiency.	Methodology limited by proprietary restrictions on some tools; lacks open-source reproducibility .
2025 [5]	Explore the evolution, architectures, and applications of AI systems in cinematic and visual storytelling content	Theoretical and empirical study of AI architectures (transformers, diffusion models, and graph-based storytelling agents). Analyzes industrial case studies like Netflix’s AI-driven recommendation-to-production pipeline.	Provides a broad yet insightful understanding of how AI impacts the entire film lifecycle — from scriptwriting to VFX.	Focuses broadly on AI evolution, offering less film-specific depth . Limited empirical validation for creative storytelling algorithms.
2024 [6]	Conduct a systematic review of AI use in digital video production, emphasizing editing, enhancement, and synthesis workflows.	Applies PRISMA-based systematic review methodology across 120+ papers from 2015–2024, focusing on AI-assisted color grading, video stabilization, and scene generation.	Identifies key research trends , frameworks, and gaps in current AI video systems. Helps establish baseline benchmarks for academic replication.	Depends heavily on secondary data; lacks primary user or creator feedback on usability of AI systems
2024 [7]	Analyze AI’s impact on media industries and film production economics.	Analytical study combining industry reports, production statistics, and AI market insights. Incorporates	Highlights key industry shifts such as automation in editing and cost reductions in pre-visualization.	Limited empirical evidence due to lack of standardized datasets on AI

		interviews with media professionals and trend analysis of adoption rates.	Maps the changing labor landscape in film production.	adoption rates.
2024 [8]	Navigate AI-driven ethical and practical challenges in filmmaking	Comparative policy analysis combined with content evaluation of AI-generated media for bias, originality, and cultural representation.	Raises ethical awareness about authorship, originality, and cultural bias. Provides framework for responsible AI use in entertainment.	Lacks practical enforcement strategies or legal precedents for AI authorship rights
2023 [9]	Generate realistic movie scripts and video scenes using LLM-guided video generation models	Combines GPT-based language models with diffusion-based text-to-video generators. Employs prompt engineering for emotional and thematic consistency.	Achieves consistency across scenes with coherent dialogues. Bridges gap between textual and visual storytelling using multimodal alignment.	Technology in early stages ; limited realism in human motion and facial expressions. Requires large-scale compute resources.
2020 [10]	Explore early applications of AI in video editing, post-production, and streaming systems.	Conceptual and technical study of early tools like Adobe Sensei, IBM Watson Media, and Google AutoML applied to video workflows.	Demonstrates foundational scope of AI in automating editing, tagging, and adaptive streaming quality	Dated dataset and limited scope of AI algorithms available in 2020. Focused on automation, not creative generation.

Table 1: Evaluation of Multi-Agent Generation Systems

2.2.Books/ Journals/ Articles referred:

1. FilmAgent:

A Multi-Agent Framework for End-to-End Film Automation in Virtual 3D Spaces is an innovative multi-agent framework designed to automate the filmmaking process in

virtual 3D environments. It simulates the roles of an entire film crew—including directors, screenwriters, actors, and cinematographers—using large language models (LLMs) to collaboratively produce a film from concept to final output. The framework supports idea development by generating structured story outlines, scriptwriting with detailed dialogues and character actions, and cinematography planning to determine optimal camera setups for each scene. To ensure high-quality output, FilmAgent leverages multi-agent collaboration strategies such as Critique-Correct-Verify and Debate-Judge, which allow agents to iteratively refine and improve their contributions. Human evaluations across 15 ideas and four key aspects showed that FilmAgent outperforms existing baselines, achieving an average score of 3.98 out of 5, demonstrating the feasibility and effectiveness of multi-agent collaboration in automated filmmaking.

2. Realistic Video Synthesis from Audio using GAN:

Realistic Video Synthesis from Audio using GAN is a system that generates realistic videos directly from audio input. The pipeline begins by processing audio through tools such as Whisper to extract speech, which is then converted into structured scripts using NLP-based approaches. These scripts guide the generation of corresponding video content that aligns with the audio input. To enhance the visual quality, the system applies Real-ESRGAN for super-resolution, resulting in high-definition and contextually accurate videos. This approach effectively combines audio processing, natural language understanding, and GAN-based visual synthesis to produce videos that are coherent, narratively aligned, and visually appealing. The system demonstrates the potential for fully automated video generation based solely on audio inputs, bridging the gap between sound and visual storytelling.

3. Preacher:

Paper-to-Video Agentic System is a novel system that converts research papers into structured video abstracts. It employs a top-down approach to first decompose, summarize, and reformulate the key content of the paper. Then, a bottom-up video generation process synthesizes diverse video segments into a coherent abstract. Preacher introduces innovations such as Key Scene Definition, which identifies important scenes to align cross-modal representations, and Progressive Chain of Thought (P-CoT), which allows granular and iterative planning to improve video coherence. The system also relies on multi-agent collaboration to handle various aspects of the paper-to-video conversion. Evaluations show that Preacher can generate high-quality video abstracts across multiple research domains, outperforming existing video generation models and demonstrating expertise in translating complex academic content into engaging visual summaries.

2.3 Patent search

2.3.1. US2025303298A1: System and Method for AI-Driven Video Generation (2025)

This patent describes an advanced system that leverages artificial intelligence for the automated generation and editing of video content. The innovation centers on a modular video generation pipeline, where input data (such as scripts, images, or audio) is processed by neural networks to create coherent video sequences. Key steps include scene identification, content segmentation, and the use of generative AI models for video synthesis and transition effects. The platform supports user customization by allowing the selection of video style, pacing, and transitions, and integrates deep learning algorithms for automatic background music and voiceover generation. This method addresses challenges in traditional video production by reducing manual labor, enhancing creative flexibility, and enabling scalable, high-quality video creation for marketing, training, and entertainment applications.

European Patent:

<https://worldwide.espacenet.com/patent/search/family/090571902/publication/US2025303298A1?q=pn%3DUS2025303298A1>

2.3.2. US12111859B2: Enterprise Generative Artificial Intelligence Architecture (2023)

This patent discloses a comprehensive architecture for enterprise-level generative artificial intelligence, including modules for video generation. The system integrates large-scale neural networks capable of synthesizing video content from text, images, or structured data inputs. It features a modular design for scalability, supporting distributed training and inference, and includes tools for content moderation, style transfer, and real-time video rendering. The architecture is designed for deployment in cloud environments, enabling businesses to automate the creation of marketing, training, and explainer videos with high efficiency and customization. The patent emphasizes secure data handling and compliance with enterprise IT standards, making it suitable for large organizations seeking to leverage generative AI for video production.

European Patent:

<https://worldwide.espacenet.com/patent/search/family/091472672/publication/US12111859B2?q=US12111859B2>

2.5 Exiting Systems:

1. BigFootVeo:

BigFootVeo is an advanced AI-powered platform that integrates **Veo3**, a state-of-the-art model designed for text-to-video and image-to-video generation. The system enables users to input descriptive text prompts or static images and automatically generates corresponding video content that aligns with the provided

input. By leveraging deep learning techniques and multimodal understanding, BigFootVeo effectively interprets context, scene composition, motion, and visual coherence to create dynamic video sequences. Its generator offers high customization potential, allowing users to control aspects such as scene transitions, lighting, and tone. This makes it a valuable tool for creators, marketers, and educators who need to produce engaging video content without traditional editing software.

Beyond simple prompt-to-video conversion, BigFootVeo provides enhanced features such as **video refinement tools, multi-scene continuity, and frame-level control**, enabling more professional output. The system also integrates AI-based post-processing techniques to ensure temporal stability and realism across frames. Through its Veo3 backbone, the platform benefits from robust motion synthesis and object consistency, producing videos that maintain visual fidelity and semantic accuracy. BigFootVeo stands out for its efficiency in bridging static visual assets with generative motion, providing an accessible yet powerful solution for automated video generation workflows.

2. OpenAI Sora:

OpenAI's **Sora** represents a major advancement in the field of **text-to-video generation**, offering users the ability to produce short, high-quality videos directly from written descriptions. Built upon OpenAI's multimodal architecture, Sora interprets natural language prompts to synthesize visually coherent, contextually accurate, and temporally consistent videos. It incorporates powerful diffusion-based and transformer-based models capable of understanding scene semantics, motion trajectories, and visual realism. Sora's design philosophy centers on making video creation as intuitive as text generation, allowing users—without technical expertise—to craft professional-looking video narratives within seconds.

In addition to raw video synthesis, Sora focuses heavily on maintaining **scene continuity, lighting realism, and object integrity** throughout the generated sequences. It can simulate camera movements, depth perception, and environmental interactions with remarkable accuracy. OpenAI has also implemented strong safety, licensing, and ethical controls to prevent misuse of generated content, ensuring responsible AI deployment. As one of the leading innovations in generative video technology, Sora exemplifies the integration of creativity and AI automation, enabling filmmakers, advertisers, and digital storytellers to accelerate the content production process while preserving artistic intent.

3. YesChat.ai:

YesChat.ai is a versatile AI platform that extends beyond conversational capabilities to offer **video generation services powered by Veo3 APIs**. It allows users to generate short video clips by providing either text prompts or visual references, which are then processed through Veo3's generation pipeline. This integration empowers users to rapidly transform abstract ideas into motion-rich video outputs. YesChat.ai focuses on delivering user-friendly interfaces that streamline the entire creative

process—from prompt input to video preview—making it accessible to both technical and non-technical audiences. Its API support also enables developers to embed video generation functionality into external applications or production pipelines.

One of YesChat.ai’s distinguishing features is its ability to provide **real-time previewing, parameter adjustments, and multi-format export options**. The platform also emphasizes high-quality rendering and maintains consistency across generated frames using Veo3’s temporal modeling. By combining conversational AI and video generation, YesChat.ai introduces an interactive way of producing multimedia assets, allowing users to iteratively refine prompts through dialogue before generating the final video. This conversational approach makes video creation more adaptive and context-aware, highlighting YesChat.ai’s unique position at the intersection of generative AI and interactive media design.

2.6. Lacuna in the existing systems:

Traditional video pre-production remains one of the most **labor-intensive and time-consuming** stages in the content creation pipeline, often resulting in significant bottlenecks for creative teams. The process of converting creative briefs into structured production plans, while simultaneously sourcing and organizing required assets, is highly fragmented and inefficient. This manual approach not only slows down project timelines but also drives up operational costs. Moreover, the lack of automation across key tasks—such as asset generation, organization, and verification—further compounds inefficiency, leaving teams to rely on repetitive, manual work that could otherwise be streamlined through AI-driven tools.

Another major challenge lies in the **high expertise barrier** associated with existing workflows. Effective use of AI and digital tools for creative production often demands advanced technical skills, placing smaller organizations and individual creators at a disadvantage. Additionally, **legal compliance** presents an ongoing concern, as manually verifying asset licenses is both complex and error-prone. This introduces potential legal risks and undermines trust in the final deliverables. Overall, the absence of automation, efficiency, and compliance mechanisms in traditional video pre-production highlights a strong need for AI-assisted systems capable of optimizing creative workflows while minimizing manual effort and human error.

2.7 Comparison of existing systems and proposed area of work:

Feature	Existing Systems	Proposed AI Agent for Video Pre-Production Planning and Asset Sourcing
Core Functionality	Focused on text-to-video or image-to-video generation, producing short clips directly from prompts	Automates the entire pre-production process including brief interpretation, planning, sourcing, and generation.

Workflow Coverage	Limited to the generation stage of video creation. No support for planning, structuring, or iteration.	Covers end-to-end pre-production — from creative brief analysis to structured storyboards and asset inventories.
Automation Level	Partial automation in generation; manual input needed for organization, sourcing, and validation.	Fully automated pipeline using collaborative AI agents for sourcing, generation, and compliance checking.
Legal & Licensing Compliance	Minimal or no built-in verification of asset licenses, increasing legal risks.	Integrates license filtering and compliance validation to ensure all sourced assets meet legal standards.
User Interaction & Collaboration	Basic user interfaces with prompt-based inputs; lacks iterative feedback or refinement features.	Includes a conversational interface for iterative refinements, version tracking, and user collaboration.
Target Audience & Accessibility	Primarily for users generating short creative clips; requires manual setup for production-level tasks.	Designed for content teams and creators seeking efficiency, scalability, and accessibility without deep technical expertise.

Table 2: Comparative Study Between Existing Systems and the Proposed AI Agent for Video Pre-Production

2.8. Focus Area:

The primary focus of this project is to **automate and optimize the video pre-production phase** using an AI-driven, agent-based architecture. Unlike existing systems that emphasize text-to-video generation, this work concentrates on the **planning, organization, and sourcing stages** that precede video creation. The system's core objective is to transform unstructured creative briefs into structured, license-compliant production kits through intelligent interpretation, shot and asset planning, and automated decision-making. By integrating **collaborative AI agents**, the platform ensures seamless coordination between brief understanding, video classification, asset sourcing or generation, and legal verification. The focus is on improving **efficiency, accessibility, and compliance** while significantly reducing the manual effort required in traditional pre-production workflows. Ultimately, the project aims to empower content creators and production teams with a scalable, user-friendly tool that bridges the gap between creative ideation and production readiness.

Chapter 3: Requirements

3.1 Proposed model

The proposed model is an agentic AI platform centered around an **Agent Orchestrator (LLM)** that manages collaborative AI agents to automate video pre-production. The workflow begins with a **Brief Interpretation Module** parsing user requirements submitted through a conversational chat interface. A **Planner Agent** then generates a detailed shot and asset list. A **Source/Generate Decider Agent** intelligently chooses whether to find assets via a **Searching Agent** (with web crawling and license filtering) or create them with a **Generation Agent** using model APIs. All data and assets are stored in PostgreSQL and vector databases, and the final output is a comprehensive **Production Kit** containing storyboards, asset inventories, and downloadable resources.

3.2 Functional Requirements

- **Brief Interpretation & Planning:** The system must parse creative briefs, classify the video type, and generate detailed shot/asset plans with timings.
- **Asset Sourcing & Generation:** It must automate sourcing assets from the web with license filtering and generate new assets (images, video, audio) via model APIs.
- **Conversational Refinement:** The system must provide a chat interface for users to make iterative refinements and view versioned differences.
- **Data Management:** It will use PostgreSQL for structured plans and a vector store for semantic asset retrieval.
- **Export Functionality:** The system must deliver a complete production kit, including storyboards (PDF) and asset lists (CSV/JSON).

3.3. Non-Functional Requirements

- **Efficiency:** Reduce pre-production time compared to manual workflows.
- **Usability:** Be intuitive for creators with minimal technical expertise.
- **Legal Compliance:** Ensure all sourced assets meet licensing requirements to minimize copyright risk.
- **Scalability:** The architecture should support diverse use cases and be built on scalable cloud infrastructure.

3.4. Hardware & Software Requirements

- **Programming Languages:** Python (for the backend), JavaScript (for the React frontend).
- **AI Frameworks:** LangChain, AutoGen, or CrewAI for orchestrating the multi-agent system.
- **Databases:** PostgreSQL for storing plans and assets, and a **Vector Store** (like Qdrant or Milvus) for semantic retrieval.

- **Web Frameworks:** **React** (frontend) and a backend orchestrated by an LLM.
- **Cloud Services:** Cloud infrastructure for hosting a web app, orchestrator, and data layers.
- **APIs:** **LLM APIs** for planning and tool calling, and vendor APIs for image/video/TTS (Text-to-Speech) generation.
- **Web Tools:** Web search and crawling tools for sourcing assets from the web.
- **Tools:** **FFmpeg** for video processing (implied by "video editing" being out of scope, but processing for exports is a possibility). The system also requires a conversational chat interface for rapid refinements and an export function for production kits.

3.5. Technology and Tools utilized

- **AI Frameworks:** An agent orchestration framework like LangChain, AutoGen, or CrewAI.
- **APIs:** LLM APIs for planning and tool-calling, along with vendor APIs for image, video, and TTS generation.
- **Web Tools:** Web search and crawling tools for asset sourcing.
- **Video Processing:** FFmpeg for potential video processing tasks.

3.6. Constraints of working

- **API Dependency:** The system's performance and cost are dependent on external, third-party LLM and media generation APIs.
- **Licensing Accuracy:** The effectiveness of legal compliance is constrained by the accuracy of the automated license filtering process.
- **Content Quality:** The quality of generated assets is limited by the capabilities of the underlying generative models.
- **Scope:** The project is strictly focused on pre-production; it does not include video editing or post-production functionalities

Chapter 4: Proposed Design

4.1 Block Diagram of the proposed system

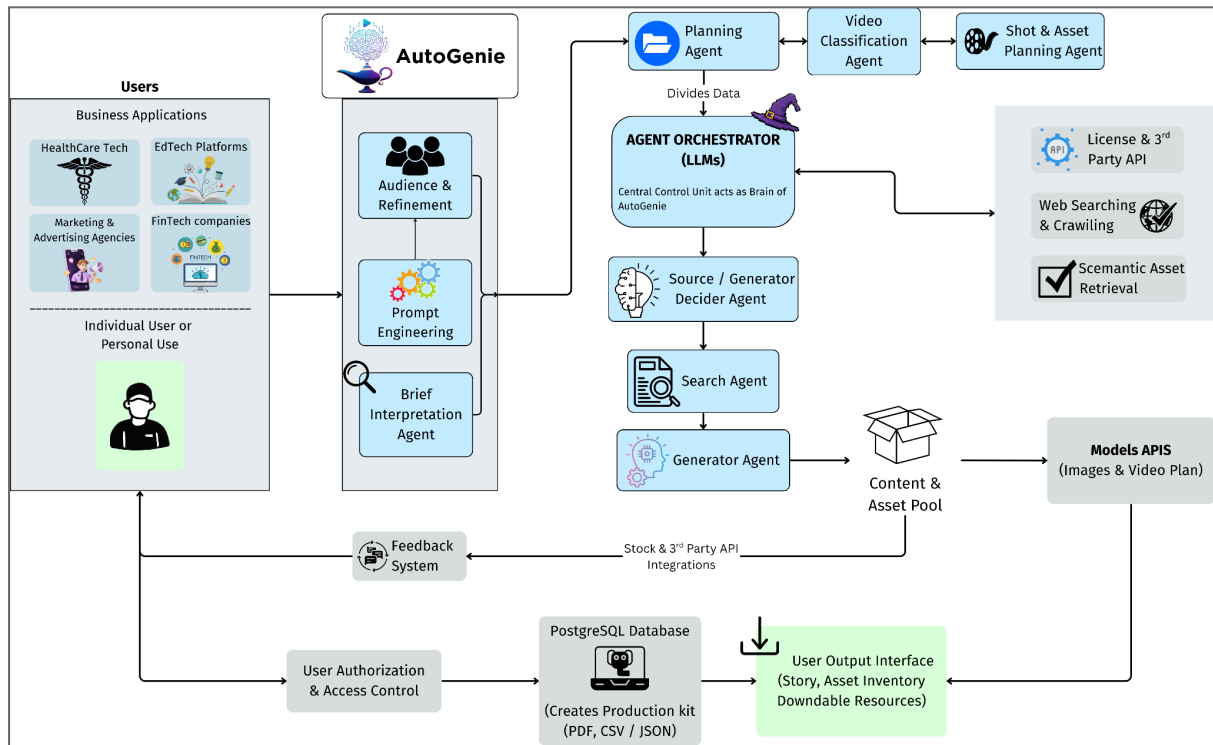


Fig 1: Detailed Agent-Based System Architecture for the AutoGenie

This diagram outlines a sophisticated AI-powered content creation platform, "AutoGenie," which begins by processing user requests from either business applications or individuals. The initial input is handled by a dedicated module where a "Brief Interpretation Agent" first understands the user's goal, "Prompt Engineering" refines it for the AI, and an "Audience & Refinement" stage tailors the objective. This structured brief is then sent to the core of the system, the "Agent Orchestrator (LLMs)," which acts as the central brain. The Orchestrator initiates the creative process by deploying a "Planning Agent" that collaborates with "Video Classification" and "Shot & Asset Planning" agents to generate a detailed, structured plan for the final content.

With a clear plan in place, the Agent Orchestrator then tasks agents to gather the necessary materials. A key "Source / Generator Decider Agent" intelligently chooses whether to find existing assets or create new ones. If searching is required, the "Search Agent" performs semantic retrieval by crawling the web and third-party APIs, ensuring all content is properly licensed. If new assets are needed, the "Generator Agent" utilizes various AI Model APIs to produce images and other media. All sourced and generated materials are collected in a "Content & Asset Pool" before being assembled and delivered to the user via the "User Output Interface" as stories, asset inventories, or other downloadable resources. The entire

process is supported by a PostgreSQL database for creating production kits and managing user authorization, while a feedback system allows for continuous improvement.

4.2 Modular diagram (with explanation)

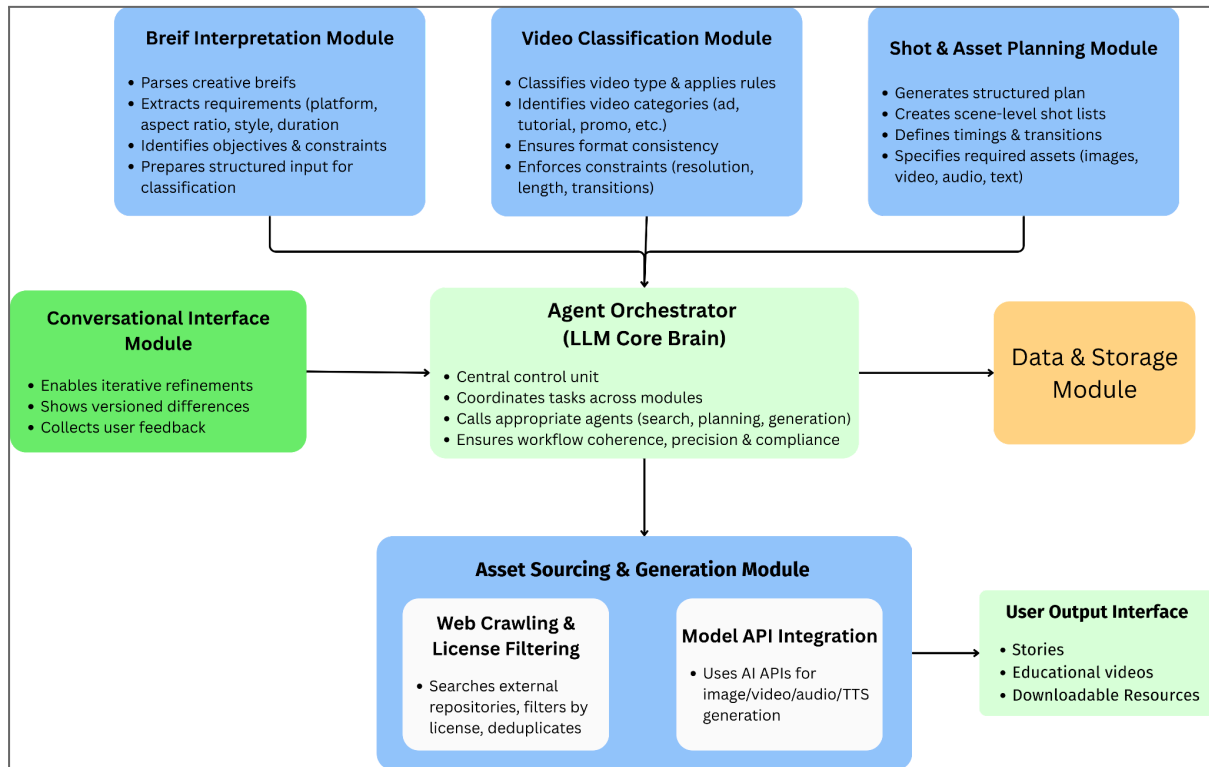


Fig 2: High-Level Modular Architecture for AutoGenie

This diagram illustrates an Agentic-AI driven automated content creation system, with the "Agent Orchestrator (LLM Core Brain)" acting as the central control unit. The process begins with the "Brief Interpretation Module," which parses a user's initial request to extract key requirements like platform, style, and objectives. This structured information is then refined through the "Conversational Interface Module," which allows for iterative user feedback and clarification, ensuring the final plan is precisely aligned with the user's vision.

Once the requirements are set, the Agent Orchestrator coordinates a series of specialized modules to plan and execute the project. The "Video Classification Module" categorizes the content type (e.g., ad, tutorial) to apply relevant rules and constraints, while the "Shot & Asset Planning Module" generates a detailed, scene-by-scene plan, specifying all necessary assets, timings, and transitions. This structured plan is then passed to the "Asset Sourcing & Generation Module," which uses web crawling to find licensed assets and AI model APIs to generate new media like images, audio, and voiceovers.

Finally, all the sourced and generated assets are compiled according to the plan, with the "Data & Storage Module" serving as the central repository for all project files and metadata. The completed content, such as educational videos or stories, is then delivered to the user.

through the "User Output Interface." This modular architecture allows for a flexible and powerful workflow, breaking down the complex task of video creation into manageable, automated steps.

4.3 Detailed Design of Detailed Process Flows for Asset Sourcing, Ingestion, and Analysis with explanation

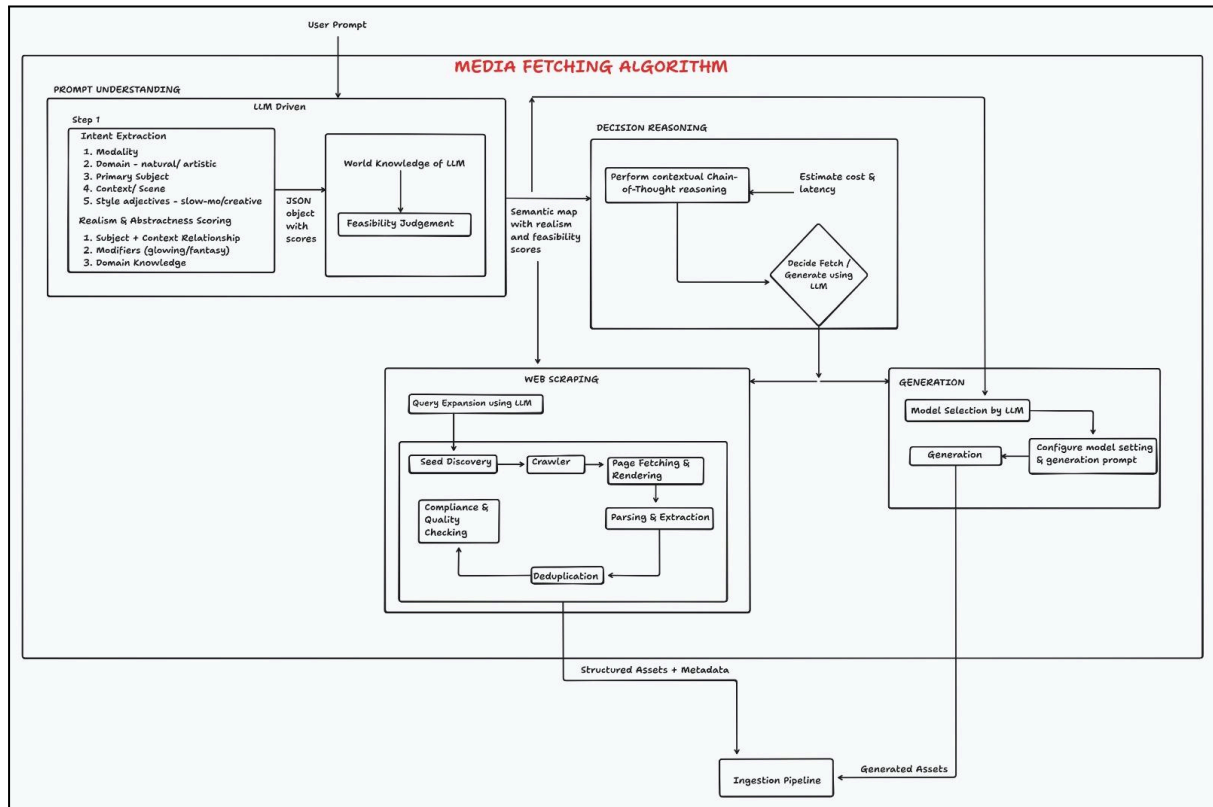


Fig 3: Detailed Diagram of the Media Fetching Algorithm

This diagram details the logic of the "Source/Generate Decider Agent", which is the core of the Asset Sourcing & Generation Module. The process begins with Prompt Understanding, where an LLM analyzes the user's prompt to extract intent, modality, style, and subject, generating a JSON object with scores. This output, combined with the LLM's world knowledge, informs a Feasibility Judgement to determine if the request is viable.

This judgment feeds into the Decision Reasoning block, which uses contextual chain-of-thought reasoning to estimate cost and latency, ultimately deciding whether to fetch (source) or generate the asset. If fetching is chosen, the Web Scraping module uses LLM-based query expansion to crawl, parse, and deduplicate assets. If generation is chosen, the Generation module selects the appropriate model and configures the generation prompt. Both pathways feed their results into the main Ingestion Pipeline for further processing.

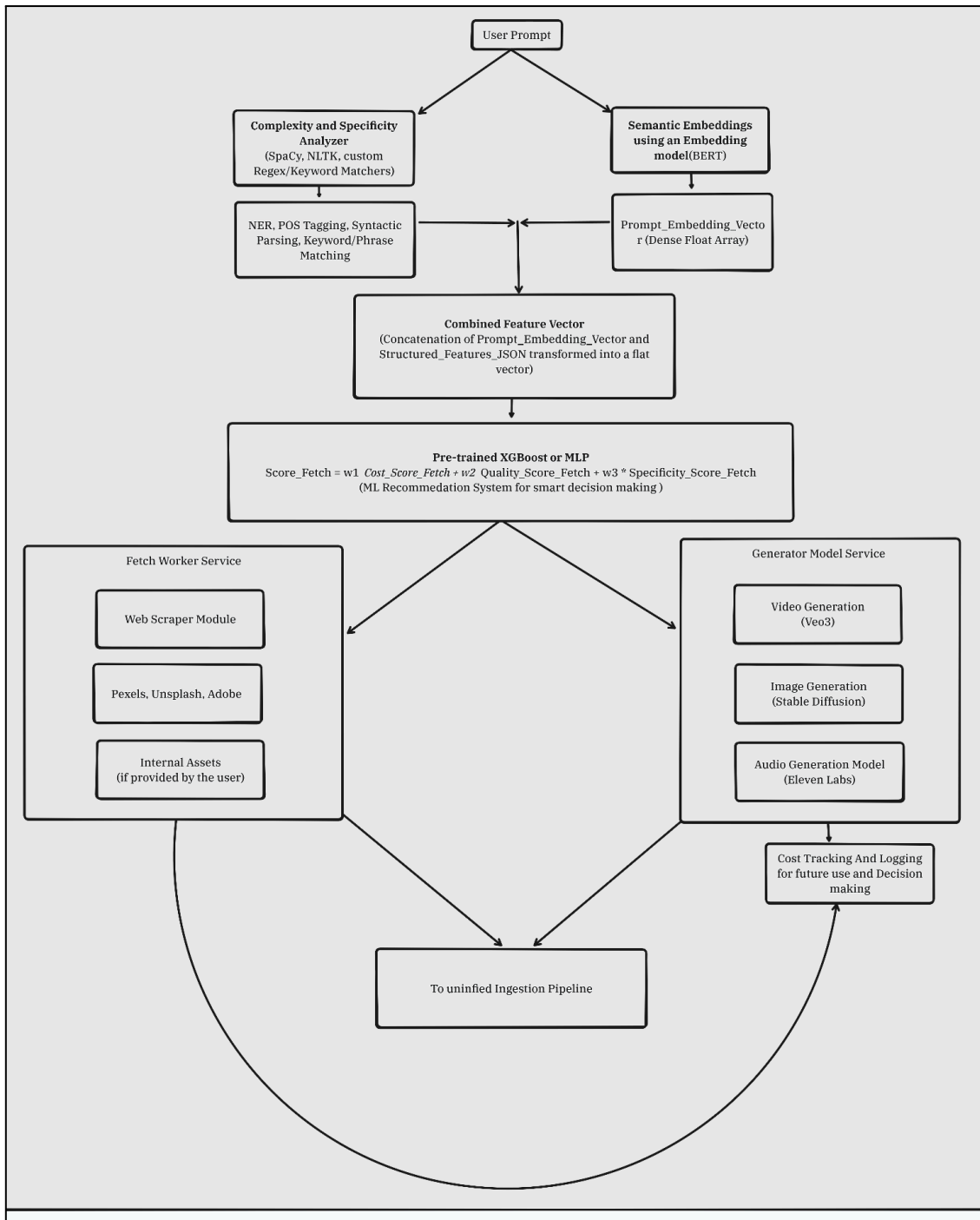


Fig 4: Machine Learning-Based App Router for Sourcing vs. Generation

This diagram presents the architecture of the "App Router," which functions as the system's primary "Source/Generate Decider Agent". This router employs a machine learning-based approach to intelligently decide whether to fetch an existing asset or generate a new one. When a user prompt is received, it undergoes parallel processing: a **Complexity and Specificity Analyzer** performs linguistic analysis (NER, POS tagging), while an **Embedding Model** creates a semantic vector representation of the prompt.

These outputs are concatenated into a **Combined Feature Vector**, which is then fed into a pre-trained model (like XGBoost or a Multi-Layer Perceptron). This model calculates weighted scores for both fetching and generating. Based on which score is higher, the task is routed to either the **Fetch Worker Service** for web scraping or to the **Generator Model Service** to create new media (video, image, or audio). Both services ultimately channel their outputs to the **Unified Ingestion Pipeline**.

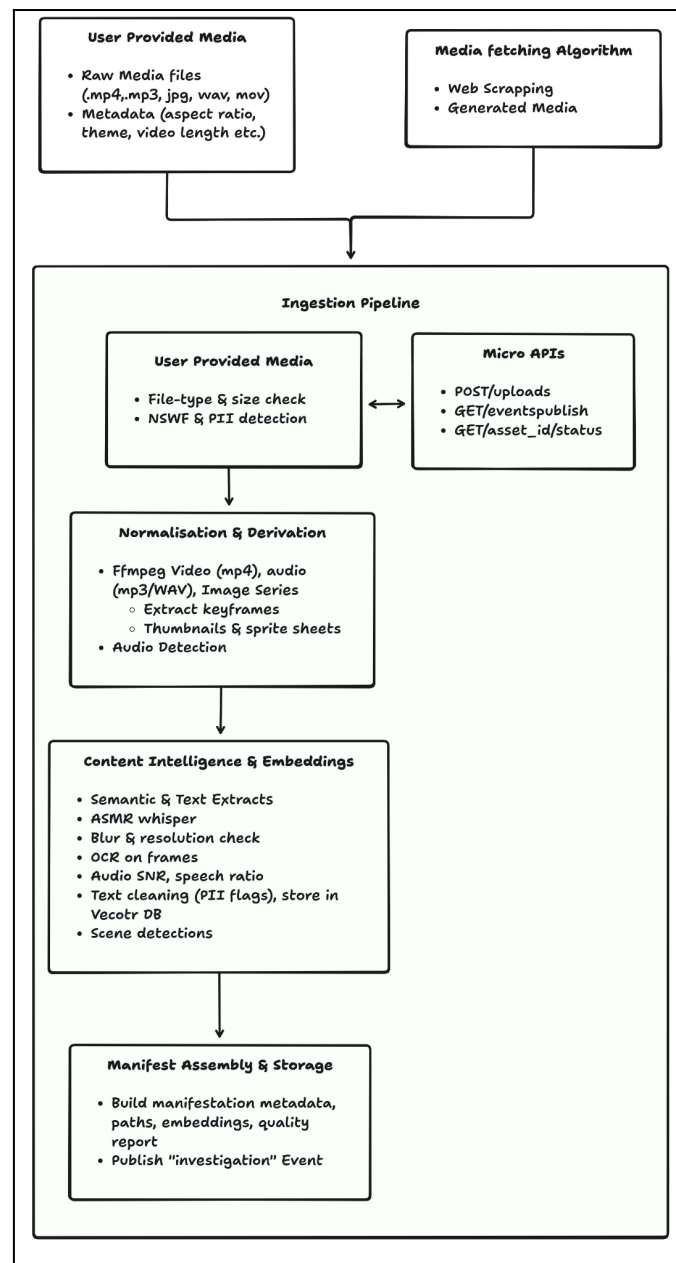


Fig 5: The Unified Asset Ingestion Pipeline

This flowchart details the **Ingestion Pipeline**, a critical component of the **Data & Storage Module**. This pipeline is responsible for processing all incoming assets, whether they are provided by the user or acquired by the **Media Fetching Algorithm**. The pipeline begins by handling the raw media files and their metadata, using micro-APIs for uploads and status tracking.

Next, the **Normalization & Derivation** step uses tools like FFmpeg to standardize file formats (e.g., to mp4, mp3) and extract key components like image series, keyframes, and audio tracks. The **Content Intelligence & Embeddings** module then performs a deep analysis, using models for ASR (e.g., Whisper), running OCR on frames, detecting scenes, and cleaning text for PII, before storing the data in a vector database. Finally, the **Manifest Assembly & Storage** step builds a comprehensive metadata manifest, generates a quality report, and publishes the processed asset for use in the project.

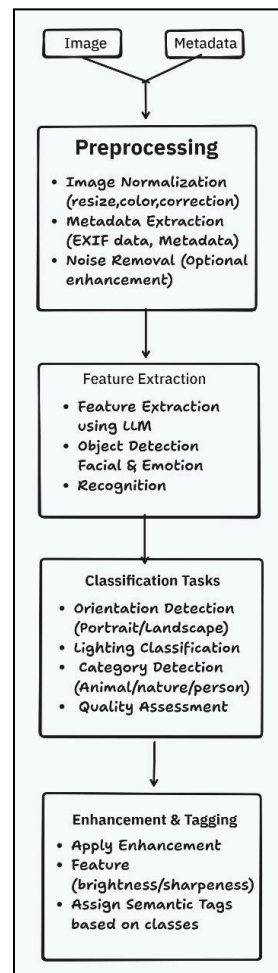


Fig 6: Image Classification and Tagging Workflow

This diagram illustrates the detailed workflow for the **Video Classification Module**, focusing on how individual images are processed after ingestion. The process begins with **Preprocessing**, where the image is normalized (e.g., resized, color-corrected) and its EXIF metadata is extracted. Following this, the **Feature Extraction** step uses a combination of an LLM, object detection, and facial/emotion recognition to understand the image's content.

These extracted features are then fed into a series of **Classification Tasks** to determine key attributes such as orientation (portrait/landscape), lighting conditions, content category (e.g., animal, nature, person), and overall quality. Finally, the **Enhancement & Tagging** module applies any necessary automated corrections (like brightness or sharpness) and assigns

semantic tags based on the classification results. This makes the asset easily searchable and ready for the **Relevance Matching** module.

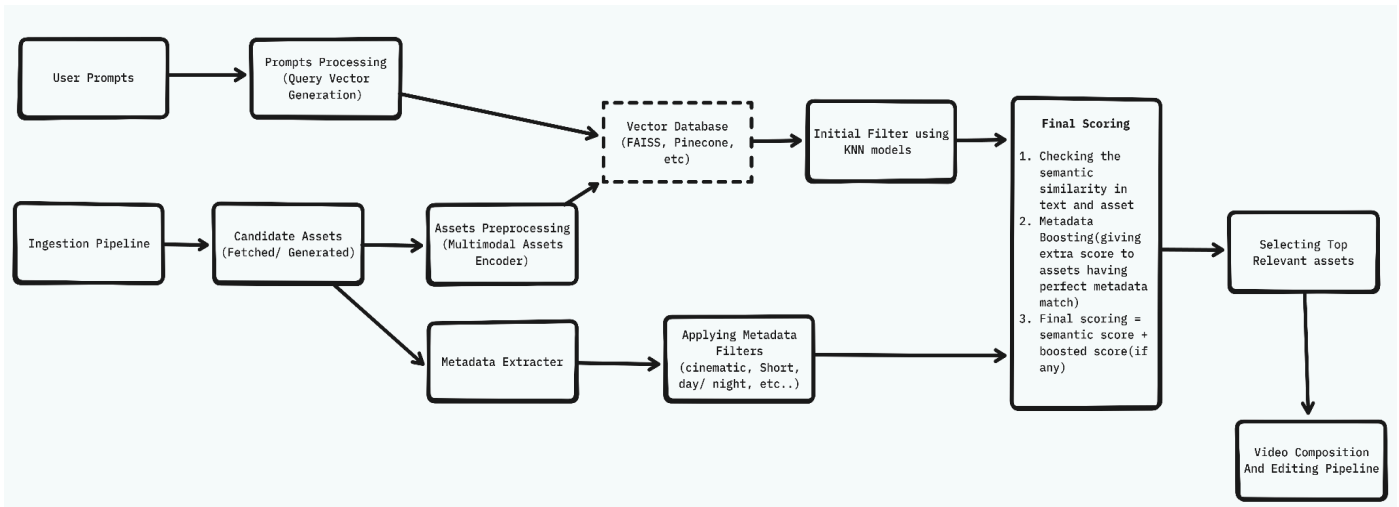


Fig 7: Relevance Matching and Final Asset Selection

This diagram outlines the **Relevance Matching** process, which ensures that the assets processed by the ingestion pipeline align with the user's original creative brief. The workflow operates on two parallel tracks: the user's prompt is processed to generate a query vector, while candidate assets from the pipeline are preprocessed to extract their own multi-modal vectors and metadata.

Both vector types are fed into a **Vector Database** (such as Qdrant or Milvus), which uses algorithms like K-Nearest Neighbors (KNN) to perform an initial similarity search. The resulting candidates then undergo **Final Scoring**, where the system checks semantic similarity, boosts scores for assets with matching metadata (e.g., shot type, time of day), and calculates a final relevance score. This allows the system to select only the top relevant assets to pass on to the final video composition plan.

4.4. Proposed algorithms

4.4.1. Agentic Multi-Agent Orchestration (LLM-based)

- **Description:** The system uses a Large Language Model (LLM) as an orchestrator to coordinate multiple specialized agents. Each agent is responsible for a distinct sub-task (e.g., planning, sourcing, generation).
- **Algorithmic Approach:**
 - **Prompt Interpretation:** The LLM parses user briefs and classifies the video type.
 - **Planning:** The Planner Agent (LLM-driven) generates a detailed shot and asset list, using chain-of-thought reasoning and prompt engineering.

- **Source/Generate Decision:** A rule-based or LLM-guided agent decides whether to source assets from the web or generate them using AI models, based on license, relevance, and quality.
- **Asset Sourcing:** The Searching Agent crawls web sources/APIs, applies license filtering, and uses semantic scoring (e.g., CLIP, DINO embeddings) to rank assets.
- **Asset Generation:** The Generation Agent calls external APIs (e.g., OpenAI, Runway) to generate images, video, or audio using prompt templates.
- **Deduplication & Embedding:** Embeddings are generated for all assets, and near-duplicates are filtered using vector similarity search.
- **Data Management:** All metadata and embeddings are stored in PostgreSQL and a vector database for efficient retrieval.

4.4.2. Retrieval-Augmented Generation (RAG) Implementation

- **Description:** RAG is suggested for improving the relevance and factuality of generated content by combining LLMs with a retrieval system.
- **Algorithmic Approach:**
 - **Retrieval:** When the LLM needs to generate a plan or asset description, it first queries a vector database (e.g., Qdrant, Milvus) containing indexed documents, previous plans, or asset metadata.
 - **Augmentation:** The retrieved context is injected into the LLM prompt, grounding the generation in real, relevant data.
 - **Generation:** The LLM then produces output (e.g., shot lists, asset prompts) that is both creative and contextually accurate.
 - **Applications:** RAG is used for plan refinement, asset description generation, and ensuring that generated content aligns with user requirements and available assets.

4.4.3. Semantic Scoring and Filtering

- **Description:** To ensure only relevant and high-quality assets are selected, the system uses semantic similarity scoring.
- **Algorithmic Approach:**
 - **Embeddings:** Assets (images, videos, audio) and user queries are embedded using models like CLIP or DINO.
 - **Similarity Calculation:** Cosine similarity is computed between asset embeddings and the query embedding.
 - **Thresholding:** Assets above a certain similarity threshold are selected; others are filtered out.

4.4.4. License Compliance Filtering

- **Description:** Automated checks ensure all sourced assets meet legal and licensing requirements.
- **Algorithmic Approach:**
 - **Metadata Extraction:** License information is parsed from asset metadata.
 - **Rule-Based Filtering:** Only assets with approved licenses (e.g., Creative Commons) are retained.

5. Conversational Refinement Loop

- **Description:** The system supports iterative plan refinement via a chat interface.
- **Algorithmic Approach:**
 - **Versioned Diffs:** Each user refinement is tracked, and differences between versions are highlighted.
 - **LLM Feedback:** The LLM suggests improvements or clarifications based on user feedback.

4.5. Project Scheduling & Tracking using Gantt Chart

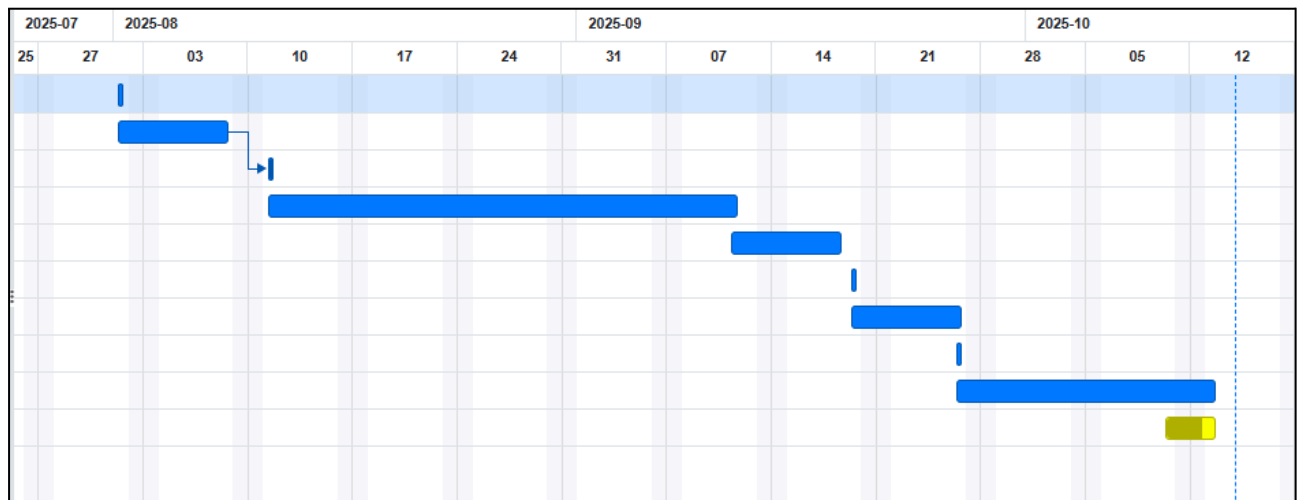


Fig 8 : Gantt Chart

This image represents the Project Scheduling and Tracking for the major project using a Gantt Chart, which visually outlines the timeline and progress of each task from August to October 2025. Each blue bar denotes a specific project phase, beginning with mentor discussion and literature survey in early August, followed by system architecture design, implementation phases, and final documentation in October. The chart helps track overlapping tasks, dependencies, and durations, giving a clear overview of how the project progresses sequentially over the three-month period.

ID ↑ :	Task Name :	Start :	End :	Duration
1	Project Allotment & Mentor Discussion	2025-08-01	2025-08-01	1 day
2	Literature Survey & Requirement Gathering	2025-08-01	2025-08-08	6 days
3	Industry Meeting 1 - Deliverable Gathering	2025-08-11	2025-08-11	1 day
4	System Architecture & Module Design	2025-08-11	2025-09-11	24 days
7	Implementation Phase 1: Asset Management	2025-09-11	2025-09-18	6 days
8	Review 1	2025-09-19	2025-09-19	1 day
9	Post-Review Refinement & Minor Bug Fixes	2025-09-19	2025-09-26	6 days
10	Industry Meeting 2 – Next Deliverable Assigned	2025-09-26	2025-09-26	1 day
11	Implementation Phase II- Generation Module	2025-09-26	2025-10-13	12 days
12	Documentation & Stage-1 Report Compilation	2025-10-10	2025-10-13	2 days

Fig 9: Tasks and Discussions Done in Gantt Chart

This image provides a tabular summary of all project tasks and discussions corresponding to the Gantt chart. It lists twelve activities with their respective start and end dates, durations, and sequence—beginning from “Project Allotment & Mentor Discussion” on August 1, 2025, and concluding with “Documentation & Stage-1 Report Compilation” on October 13, 2025. This structured table highlights key milestones such as industry meetings, review sessions, and implementation phases, enabling efficient monitoring of deadlines and ensuring the project stays on schedule.

Chapter 5: Results and Discussions

For the prompt “**girl sitting near seashore**”, the multimodal retrieval pipeline successfully identified and ranked a set of **semantically relevant images and videos** from the Pexels database. Using **OpenAI LLM** for query planning and refinement, and **OpenCLIP (ViT-L/14)** for visual-text similarity scoring, the system iteratively filtered out unrelated content and converged on the most contextually accurate assets.

The final ranked outputs included images and videos depicting scenes such as:

- a young girl sitting on a rocky beach or by the ocean,
- a child or woman seated near the waterline,
- individuals relaxing or playing by the seaside during sunny weather.

```
{
  "prompt": "girl sitting near seashore",
  "global_mood": "Calm",
  "generated_queries": [
    "girl at seashore",
    "girl sitting beach",
    "seashore girl photo",
    "girl by ocean",
    "beach scene girl"
  ],
  "results": {
    "girl at seashore": {
      "query_mood": "Calm",
      "pexels": {
        "photos": [
          {
            "id": "photo_323244",
            "url": "https://images.pexels.com/photos/323244/pexels-photo-323244.jpeg",
            "width": 1280,
            "height": 1920,
            "photographer": "Summer Stock",
            "provider": "pexels"
          },
          {
            "id": "photo_825904",
            "url": "https://images.pexels.com/photos/825904/pexels-photo-825904.jpeg",
            "width": 5600,
            "height": 3575,
            "photographer": "Andrea Piacquadio",

```

The **top similarity scores ranged between 0.21 and 0.25**, indicating strong alignment between the visual semantics of the retrieved assets and the textual concept of “*girl sitting near seashore.*”

These results confirm that the pipeline effectively captures the **semantic meaning, environment, and subject context** of natural language prompts through combined **language–vision embedding analysis**.

```
Enter user prompt: girl sitting near seashore
LLM initial plan: {'clean_prompt': 'girl sitting near seashore', 'num_images': 5, 'num_videos': 2, 'num_audio': 0, 'notes': 'The prompt'}
[info] Loading ViT-L-14 (OpenCLIP) on cpu – this may download >1.7GB if not cached.
Fetched initial 7 assets.
```

```
=== Iteration 1 === total candidates so far: 7
Embedding 7 items using ViT-L-14...
Found 6 assets with score >= 0.2 (threshold).
[info] Requesting query refinements from LLM ...
Refiner action: refine_query
Fetched 3 for query: young girl sitting near seashore
Fetched 4 for query: child sitting on the beach
Fetched 5 for query: girl playing by the water
Fetched 2 for query: girl sitting on the sand
```

```
=== Iteration 2 === total candidates so far: 21
Embedding 21 items using ViT-L-14...
Found 13 assets with score >= 0.2 (threshold).
Target satisfied: 13 >= 7. Stopping iterations.
```

Final top results (ViT-L-14 scores):

1. [image] pexels id=8510861 score=0.2514
title: A young blonde girl in swimwear sits on rocky beach seashore under the summer sun.
url : <https://images.pexels.com/photos/8510861/pexels-photo-8510861.jpeg>
2. [video] pexels id=1093667 score=0.2513
title: Peter Fowler
url : https://videos.pexels.com/video-files/1093667/1093667-hd_1920_1080_30fps.mp4
3. [image] pexels id=7849991 score=0.2496
title: Young girl in orange beach chair enjoying sunny day by the ocean.
url : <https://images.pexels.com/photos/7849991/pexels-photo-7849991.jpeg>
4. [image] pexels id=2972971 score=0.2381
title: A woman in a hat sits on a bench overlooking the ocean, enjoying a serene summer day.
url : <https://images.pexels.com/photos/2972971/pexels-photo-2972971.jpeg>
5. [image] pexels id=8510862 score=0.2377
title: Blonde child in a red swimsuit sitting on rocky beach, gazing at the sky.
url : <https://images.pexels.com/photos/8510862/pexels-photo-8510862.jpeg>
6. [video] pexels id=5444730 score=0.2276
title: Honye Sanges
url : https://videos.pexels.com/video-files/5444730/5444730-hd_1920_1080_24fps.mp4
7. [image] pexels id=1126327 score=0.2264
title: A young woman sitting on a sea wall with her bicycle by the ocean, enjoying the view.
url : <https://images.pexels.com/photos/1126327/pexels-photo-1126327.jpeg>
8. [image] pexels id=7863491 score=0.2257
title: Young girl in sun hat playing with sand on a sunny beach day.

Chapter 6: Plan of action for the next semester

The primary focus for the next semester will be the incremental development of an intelligent asset sourcing workflow using agentic AI, as envisioned in the future implementation diagram. The following actionable steps and milestones are planned:

1. Implement Unstructured Prompt Processing

- Develop an interface to receive unstructured user prompts for asset requirements.
- Integrate a Large Language Model (LLM) to analyze the prompt for context, required assets, asset quantity, and asset types.

2. Query Generation and Asset Fetching

- Design modules for constructing asset-specific queries to fetch images, videos, or sounds from sources like Unsplash, Pexels, and Freesound.
- Build connectors/APIs for automated crawling of these repositories.

3. Asset Relevance Classification

- Develop models or scripts that parse the fetched assets using alt text and image parsing to compute a relevance score for each asset.
- Design and implement a rating and thresholding system for asset filtering.

4. Automated Feedback Loop

- Fine-tune the asset-rating and filtering thresholds based on iterative feedback and testing with real prompts.
- Integrate a continuous learning loop for adjusting parameters to enhance relevance and quality.

5. Output Generation

- Structure the output to present only highly relevant assets, along with their scores and descriptive metadata to the user.
- Ensure the system can generate comprehensive reports of asset curation for production readiness.

6. Testing and Evaluation

- Conduct rigorous unit and integration testing for each workflow stage.
- Prepare evaluation criteria and collect performance metrics to document improvements in asset sourcing efficiency and precision.

7. Documentation and Reporting

- Maintain comprehensive documentation of each implemented feature and workflow.
- Prepare interim and final reports for the semester summarizing achievements, challenges, and updated plans.

Chapter 7: Conclusion

This project aims to simplify and accelerate the often time-consuming pre-production phase of video creation. By combining the strengths of AI-driven planning, automated sourcing, and safe content generation, the system acts as a supportive co-pilot for content creators and marketing teams. The solution not only improves efficiency but also ensures legal compliance, quality assurance, and creative flexibility. Through a conversational interface, users can iteratively refine their video plans, making the process more interactive and less overwhelming. In the long run, such a human-AI integrated system has the potential to become a standard tool in digital content workflows, bridging the gap between raw ideas and production-ready resources.

Chapter 8: References

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9. Appendix

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